

3rd year

Signals & Systems Lab

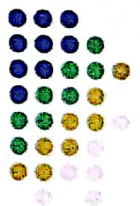
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مؤخره

Lab (10) Deconvolution

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Deconvolution



What is Deconvolution?

A mathematical inverse operation of convolution to recover the original input signal from the output of a convolution process

ترجع الإشارة الأصلية

If $y = x * h$ (convolution), then $x = \text{deconv}(y, h)$

$$y(t) = x(t) * h(t) \rightarrow \text{Convolution}$$

السواء

الدمج

$$x(t) = y(t) \otimes h(t) \rightarrow \text{Deconvolution (where } \otimes \text{ represents deconvolution)}$$



Deconvolution

The output signal must be the first input argument of the command deconv.

x=deconv(y,h)

or

h=deconv(y,x)

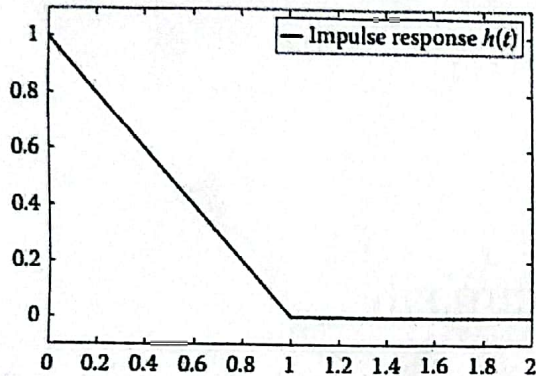
→ Inverse (reverse) operation

hh=deconv(y,x)*(1/step);

plot(t, hh)

ylim([-1 1.1]);

legend('impulse response h(t)')



The impulse response $h(t)$ is computed by multiplying the outcome of the command deconv by the quantity $(1/\text{step})$ as deconvolution is the inverse operation of convolution.

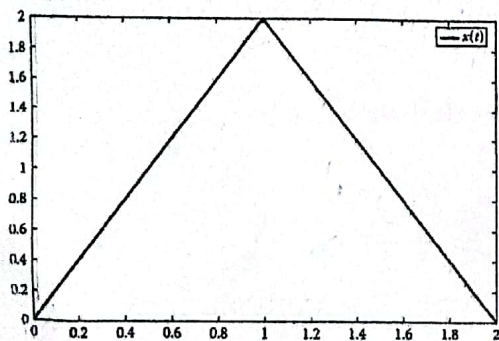
$$hh(t) = h(t) = \begin{cases} 1-t, & 0 \leq t \leq 1 \\ 0, & 1 < t \leq 2 \end{cases}$$



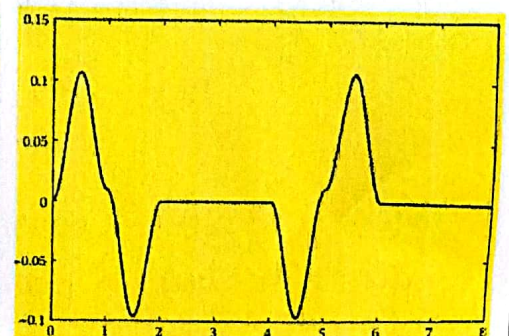
Convolution Examples

Example

Suppose that a system is described by the impulse response $h(t) = \cos(2\pi t)(u(t) - u(t-4))$. Compute and plot the response of the system to the input signal that is shown in the figure below.



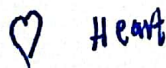
```
t1=0:.01:1;
t2=1.01:.01:2;
t3=2.01:.01:4;
x1=2*t1;
x2=4-2*t2;
x3=zeros(size(t3));
x=[x1 x2 x3];
t=[t1 t2 t3];
h=cos(2*pi*t);
y=conv(x,h)*.01;
plot(0:.01:8,y)
```



$$x(t) = \begin{cases} 2t, & 0 \leq t \leq 1 \\ 4-2t, & 1 < t \leq 2 \\ 0, & 2 < t \leq 4 \end{cases}$$



Heart



Heart



Deconvolution Examples

$$x(t) = \begin{cases} 2t, & 0 \leq t \leq 1 \\ 4 - 2t, & 1 < t \leq 2 \\ 0, & 2 < t \leq 4 \end{cases}$$

Example: Suppose that a system output $y(t)$ is obtained from a known input $x(t)$ and unknown impulse response $h(t)$. Given the same input signal $x(t)$ shown below, use deconvolution to estimate the impulse response $h(t)$ from the measured output $y(t)$.

```
t = 0:0.01:8;  
t1 = 0:0.01:1;  
t2 = 1.01:0.01:2;  
t3 = 2.01:0.01:4;  
x1 = 2*t1;  
x2 = 4 - 2*t2;  
x3 = zeros(size(t3));  
x = [x1 x2 x3];  
h = cos(2*pi*(t));  
y = conv(x, h);  
[h, r] = deconv(y, x);  
plot(t, h)
```

